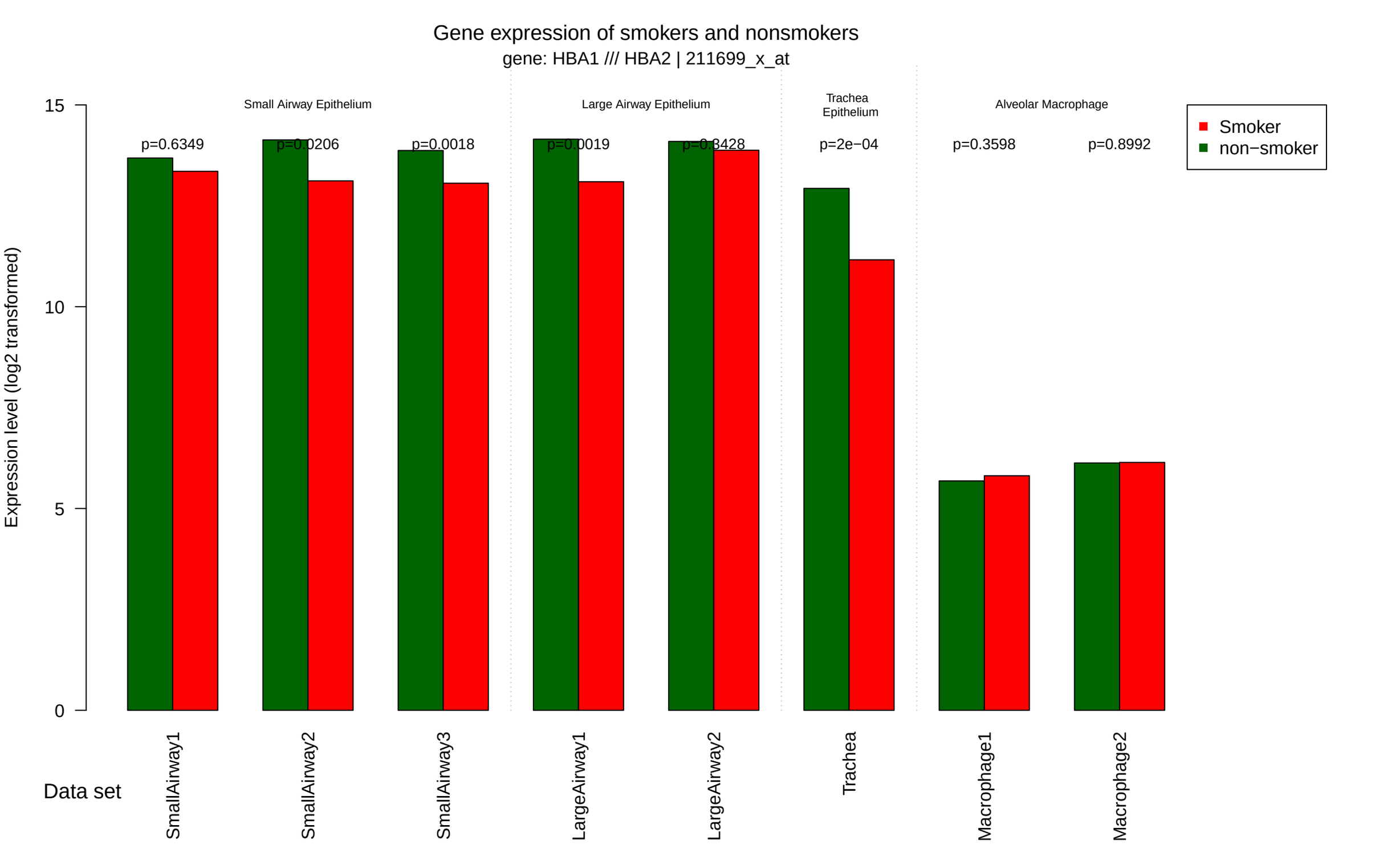
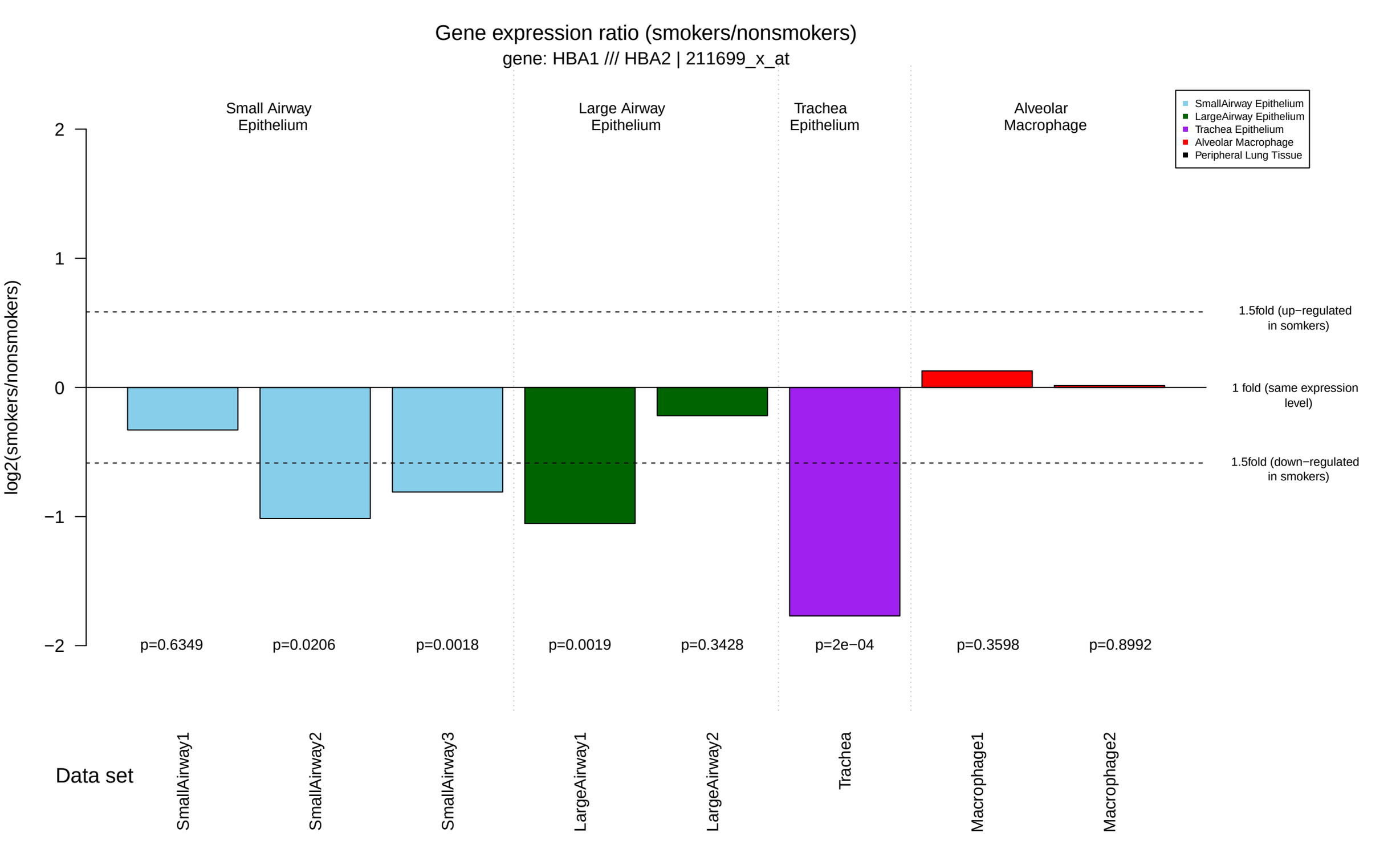
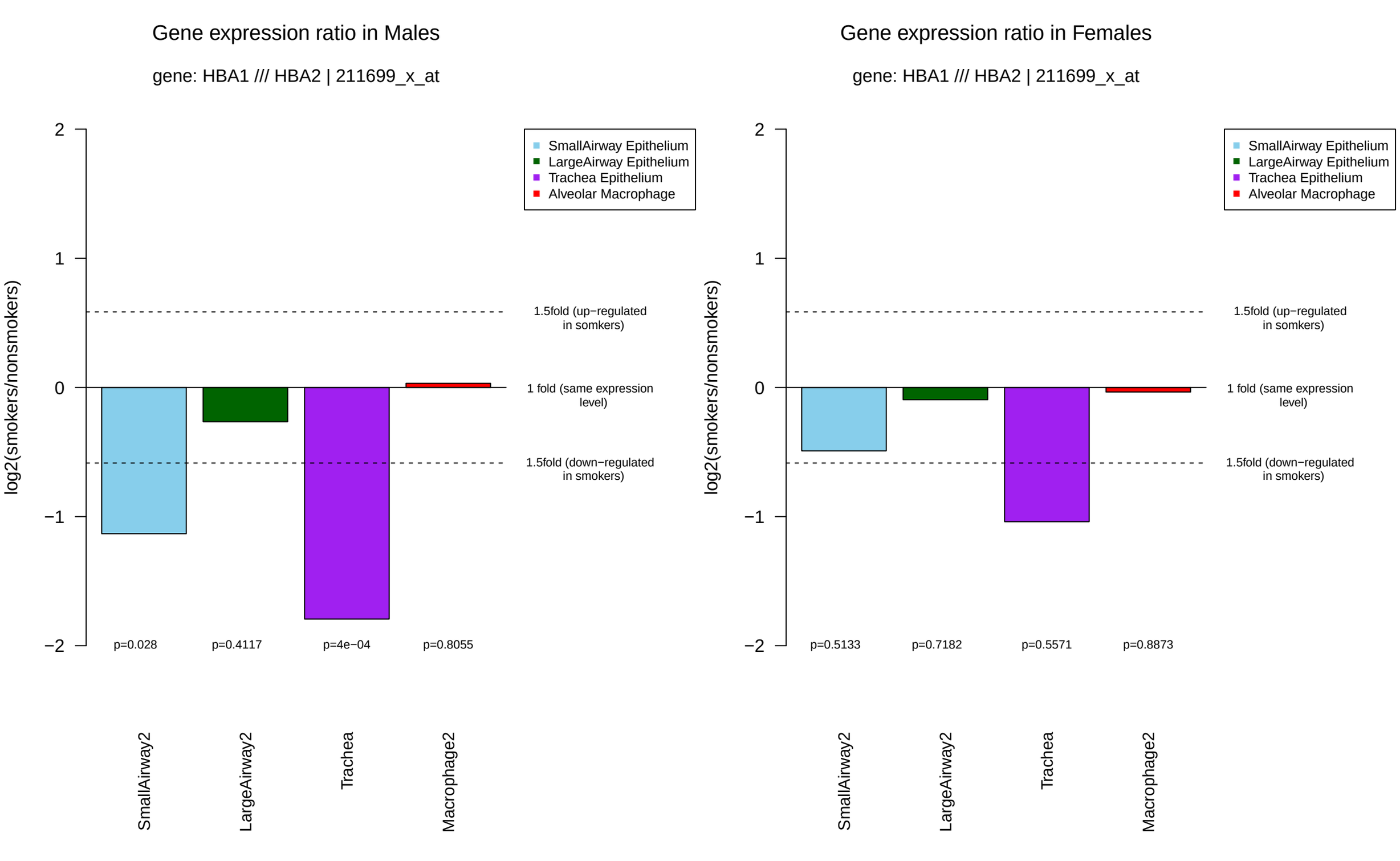
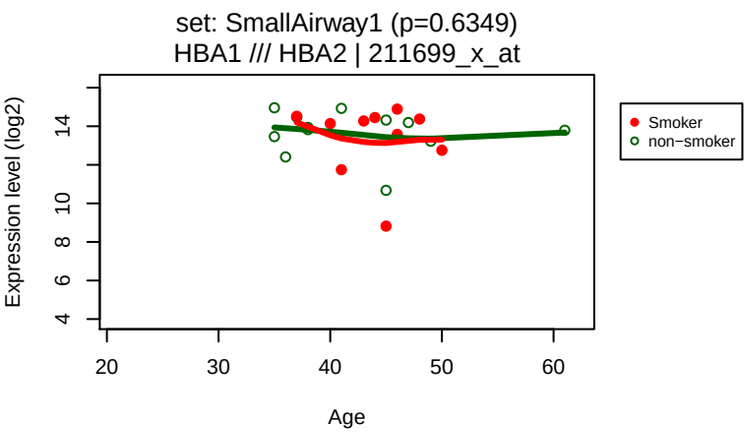
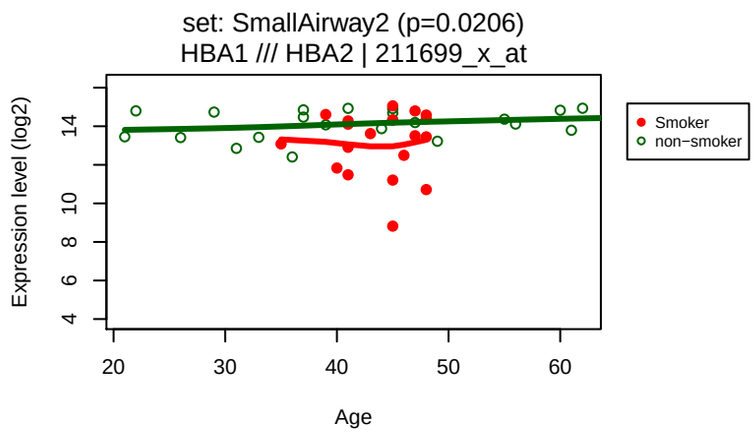
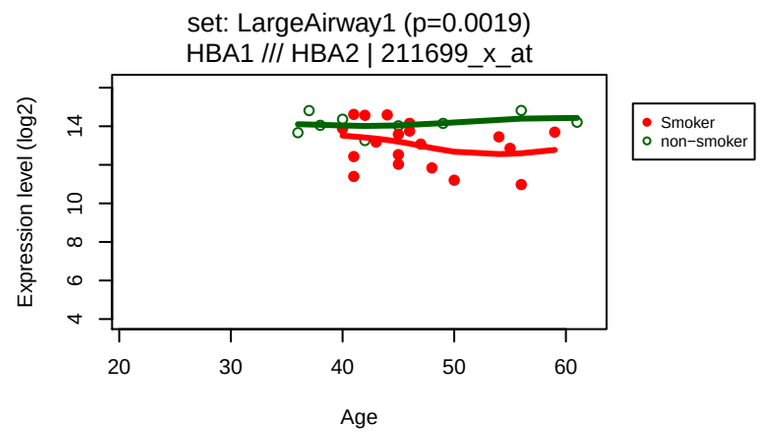

Gene expression of smokers and nonsmokers
gene: HBA1 /// HBA2 | 211699_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6349
p=0.0206
p=0.0018
p=0.0019
p=0.3428
p=2e−04
p=0.3598
p=0.8992
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

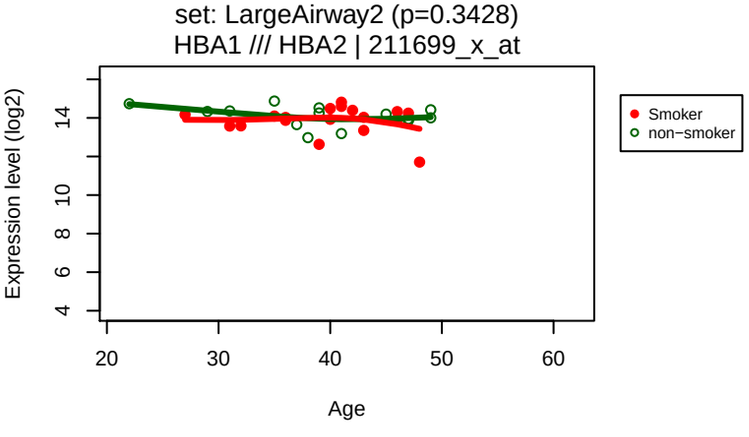

Gene expression ratio (smokers/nonsmokers)
gene: HBA1 /// HBA2 | 211699_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6349
p=0.0206
p=0.0018
p=0.0019
p=0.3428
p=2e−04
p=0.3598
p=0.8992
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

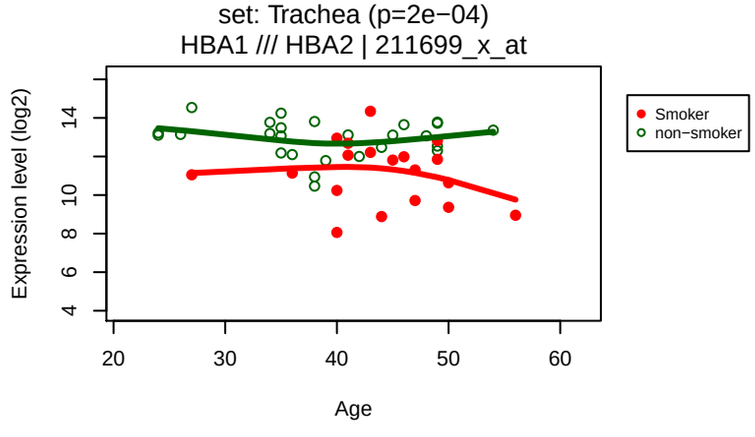

Gene expression ratio in Males
gene: HBA1 /// HBA2 | 211699_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.028
p=0.4117
p=4e−04
p=0.8055
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HBA1 /// HBA2 | 211699_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5133
p=0.7182
p=0.5571
p=0.8873
SmallAirway2
LargeAirway2
Trachea
Macrophage2

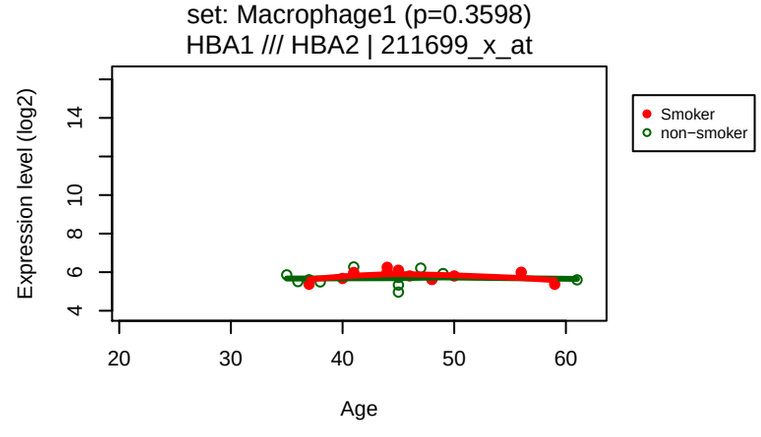

set: SmallAirway1 (p=0.6349)
HBA1 /// HBA2 | 211699_x_at
Expression level (log2)
Age
Smoker
non−smoker

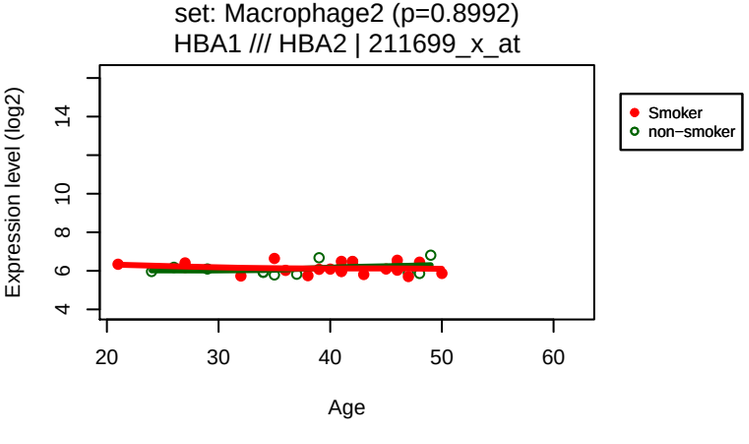

set: SmallAirway2 (p=0.0206)
HBA1 /// HBA2 | 211699_x_at
Expression level (log2)
Age
Smoker
non−smoker

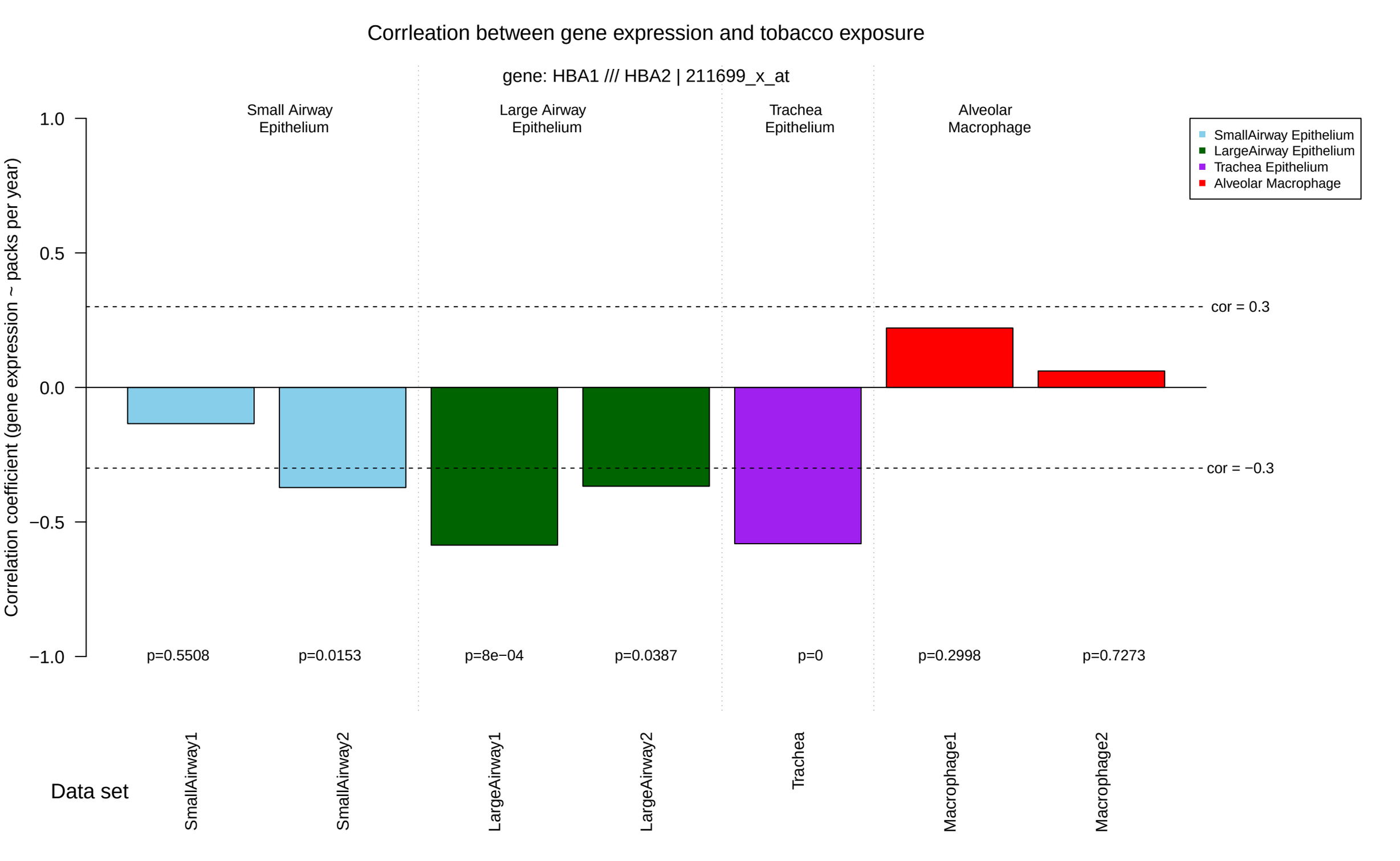

set: LargeAirway1 (p=0.0019)
HBA1 /// HBA2 | 211699_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3428)
HBA1 /// HBA2 | 211699_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=2e−04)
HBA1 /// HBA2 | 211699_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3598)
HBA1 /// HBA2 | 211699_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8992)
HBA1 /// HBA2 | 211699_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HBA1 /// HBA2 | 211699_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5508
p=0.0153
p=8e−04
p=0.0387
p=0
p=0.2998
p=0.7273
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2